

ÖZYEĞİN UNIVERSITY
Graduate School of Science and Engineering

**Design and Experimental Characterization of
a QCL-Based QEPAS Sensor for Trace
Nitrous Oxide (N₂O) Detection**

by

Burak Arayıt

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Approval

This thesis titled “**Design and Experimental Characterization of a QCL-Based QEPAS Sensor for Trace Nitrous Oxide (N₂O) Detection**” and submitted by **Burak Arayıt** in partial fulfillment of the requirements for the degree of Master of Science in Electrical and Electronics Engineering has been examined and approved by the committee below.

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Declaration

I hereby declare that this thesis is my own work, that it has been produced in accordance with academic rules and ethical conduct, and that it has not been submitted, in whole or in part, for any other degree or qualification. I have fully cited and referenced all material and results that are not original to this work.

Name, Last name: Burak Arayıt

Signature: _____

Abstract

Write a self-contained summary of your thesis here (typically 150–300 words): the problem you address, your approach/method, the key results, and what they mean. The abstract should read on its own, without reference to the rest of the document, and should avoid citations, figures, and undefined abbreviations.

Keywords: quartz-enhanced photoacoustic spectroscopy (QEPAS), nitrous oxide, quantum cascade laser, pulsed QCL, trace gas detection, amplitude modulation, quartz tuning fork

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Chapter 1

Introduction

1.1 Motivation

Nitrous oxide (N_2O) is one of the most important long-lived greenhouse gases. Its global-warming potential is roughly 270 times that of CO_2 over a 100-year horizon, and it is currently among the dominant ozone-depleting gases emitted by human activity. **[TODO: Add exact GWP figure and cite (e.g. IPCC AR6) and the dominant-ODS claim (Ravishankara *et al.*, 2009).]** The main anthropogenic sources are agriculture and soil management, combustion, and industrial processes, while N_2O also appears in medical and breath-analysis contexts. Monitoring N_2O at trace (sub-ppm) levels therefore matters for climate science, environmental regulation, and health.

Detecting such low concentrations demands sensors that are sensitive, selective, fast, and ideally compact. Laser absorption spectroscopy meets these needs, and among its variants *quartz-enhanced photoacoustic spectroscopy* (QEPAS) is particularly attractive: it uses a millimetre-scale quartz tuning fork (QTF) as a sharply resonant acoustic transducer instead of a microphone, giving an immunity to environmental acoustic noise, a very small required gas volume, and a wide dynamic range [1], [2]. Combined with a quantum cascade laser (QCL) emitting on a strong N_2O absorption band in the mid-infrared, QEPAS is well suited to compact N_2O sensing.

The broader aim of this line of work is the optical detection of trace volatile organic compounds (VOCs), such as those emitted by plants. N_2O is used here as a well-characterized reference analyte: it has strong, well-documented absorption lines that make it ideal for building and benchmarking the sensor before moving to more demanding targets.

1.2 Problem Statement and Objectives

This thesis concerns the design, construction, and experimental characterization of a QCL-based QEPAS sensor, operated with a *pulsed* QCL in amplitude-modulation mode, and its validation on N_2O . The specific objectives are to:

- assemble and align the optical, acoustic, and electronic subsystems of the QEPAS sensor;

- characterize the key components — the quartz tuning fork / acoustic detection module (ADM), the lock-in detection chain, and the pulsed QCL;
- establish an optical power budget for the beam-delivery path;
- select a suitable N_2O absorption line and demonstrate detection, including a concentration calibration; and
- quantify the sensor's performance: sensitivity (normalized noise-equivalent absorption), minimum detectable concentration, and stability via Allan-deviation analysis.

1.3 Contributions

The main contributions of this thesis are:

- a working pulsed-QCL QEPAS sensor, designed and built as part of this work;
- a thorough experimental characterization of its components and overall detection performance; and
- a validated N_2O measurement that serves as a foundation for future trace-VOC detection.

1.4 Thesis Outline

Chapter 2 reviews photoacoustic spectroscopy, the QEPAS technique, and the spectroscopy of N_2O . Chapter 3 describes the sensor design, the pulsed amplitude-modulation detection scheme, and the experimental setup. Chapter 4 presents the component characterization and N_2O detection results. Chapter 5 concludes and outlines future work toward trace-VOC sensing.

Chapter 2

Background and Related Work

This chapter provides the background needed to follow the rest of the thesis: the principle of photoacoustic spectroscopy, the QEPAS technique and its transducer, the detection scheme used with a pulsed laser, the figures of merit used to compare sensors, and the spectroscopy of N_2O .

2.1 Photoacoustic Spectroscopy

Explain the photoacoustic effect: modulated optical absorption heats the gas periodically, generating a pressure (acoustic) wave that a transducer detects. Note that the signal is proportional to the absorbed optical power, so it scales with laser power and with analyte concentration.

2.2 Quartz-Enhanced Photoacoustic Spectroscopy (QEPAS)

Describe how QEPAS replaces the microphone with a quartz tuning fork acting as a resonant piezoelectric transducer [1], optionally with an acoustic micro-resonator (the ADM). Discuss the high quality factor Q , the narrow resonance at f_0 , and why this gives strong noise immunity and a small sampling volume [2].

2.3 Detection Scheme: Amplitude Modulation with a Pulsed QCL

For a continuous-wave laser, wavelength modulation with second-harmonic (2f) lock-in detection is common. With a *pulsed* QCL, as used in this thesis, the laser intensity is instead modulated at the tuning-fork resonance f_0 (amplitude modulation), and the photoacoustic signal is recovered with a lock-in amplifier at the fundamental (1f). The absorption line is scanned by slowly tuning the laser wavelength via temperature and/or drive current. Explain the trade-offs of pulsed operation (e.g. spectral broadening, duty cycle) relative to CW.

2.4 Figures of Merit

Define the metrics used later: signal-to-noise ratio, the normalized noise-equivalent absorption (NNEA, in $\text{cm}^{-1} \text{W Hz}^{-1/2}$), minimum detectable concentration, and the use of the Allan deviation to find the optimal integration time.

2.5 Spectroscopy of Nitrous Oxide

N_2O has its strongest fundamental (ν_3) near $4.5 \mu\text{m}$, but the tunable quantum cascade laser used in this work (Daylight Solutions MIRcat-2300) covers $1111/\text{cm}$ to $2000/\text{cm}$ ($5 \mu\text{m}$ to $9 \mu\text{m}$). The accessible N_2O feature is therefore the ν_1 fundamental band near $7.7 \mu\text{m}$; the line used here is at $1297.8/\text{cm}$, with a secondary line at $1271.5/\text{cm}$. Line selection is made with the HITRAN database [3]. A key practical issue in this region is strong interference from water vapour (H_2O), which has many lines across the $1111/\text{cm}$ to $2000/\text{cm}$ window (demonstrated in Chapter 4). Nitrogen, used as the buffer gas, is essentially non-absorbing here.

2.6 Related Work

Briefly survey prior QEPAS sensors — especially N_2O and other QCL-based systems — and state how this work relates to and differs from them.

Chapter 3

Sensor Design and Experimental Setup

This chapter describes the QEPAS sensor built in this work: the pulsed quantum cascade laser source, the acoustic detection module, the amplitude-modulation detection and lock-in recovery scheme, the beam-delivery optics and alignment procedure, the optical power budget, and the gas-handling system.

3.1 System Overview

The sensor follows a standard QEPAS architecture (Figure 3.1). A pulsed, tunable quantum cascade laser (QCL) is tuned onto a N_2O absorption line and pulsed at the resonance frequency of a quartz tuning fork (QTF) housed in a commercial acoustic detection module (ADM). The beam is steered and focused between the QTF prongs by a two-mirror system. Periodic optical absorption by N_2O generates an acoustic wave that drives the QTF at resonance; the resulting piezoelectric current is amplified by the ADM's integrated preamplifier and demodulated by a lock-in amplifier referenced to the laser pulse train. A mass-flow / pressure system sets the N_2O concentration and chamber pressure. The main parameters are listed in Table 3.1.

QEPAS setup schematic (QCL $\rightarrow$ steering mirror $\rightarrow$ off-axis parabolic mirror $\rightarrow$ ADM; ADM preamp $\rightarrow$ lock-in; gas-handling rig). Replace with your block diagram / “QEPAS setup schematic” figure.

Figure 3.1. Schematic of the pulsed-QCL QEPAS sensor.

Table 3.1. Main system parameters.

Subsystem	Parameter	Value
QCL	Source	Daylight Solutions MIRcat-2300
	Tuning range	1111/cm to 2000/cm (5 μm to 9 μm)
	N ₂ O line (primary)	1297.8/cm ($\approx$ 7.7 μm)
	Pulse width	500 ns to 1000 ns
	Repetition / mod. freq.	$f_0 = 12\,458.8$ Hz (external trigger)
ADM	Module	Thorlabs ADM01
	QTF resonance f_0	12 458.8 Hz
	Windows	BaF ₂ , wedged, $\approx$ 2.5 mm
Optics	Focusing mirror	gold OAP, MPD149-M01, RFL 101.6 mm
	Focused beam waist	$\approx$ 100 μm
	Two-window transmission	87.2 %
Lock-in	Amplifier	EG&G 5110
Gas	Chamber pressure	500 Torr to 600 Torr
	N ₂ O concentration	50 ppm to 100 ppm

3.2 Excitation Source: Pulsed Quantum Cascade Laser

The excitation source is a Daylight Solutions MIRcat-2300 tunable QCL, capable of emitting between 1111/cm and 2000/cm (5 μm to 9 μm). It is operated in pulsed mode and tuned onto the N₂O ν_1 line at 1297.8/cm. The laser is driven in *external-trigger* mode: a waveform generator (Rigol DG922 Pro) supplies a 5 V (peak-to-peak) TTL pulse train at the QTF resonance frequency $f_0 = 12\,458.8$ Hz to the QCL TRIG IN input (high impedance, pulse mode), so that one laser pulse is emitted per QTF oscillation period. This constitutes the amplitude modulation of the optical power at f_0 . Average optical power is monitored with an Ophir Nova II power meter.

3.3 Acoustic Detection Module

The transducer is a Thorlabs ADM01 acoustic detection module, comprising a quartz tuning fork with an acoustic micro-resonator and an integrated transimpedance preamplifier (powered at ± 12 V from a Keysight E361A supply). The module has wedged, uncoated BaF₂ windows of ≈ 2.5 mm thickness. The measured fork resonance is $f_0 = 12\,458.8$ Hz (Section 4.1). The ADM is mounted on a five-axis translation stage (Thorlabs PY005) to allow the beam to be centred between the fork prongs,

whose gap is ≈ 0.8 mm.

3.4 Amplitude-Modulation Detection and Lock-In Recovery

Because the QCL is pulsed, detection uses amplitude modulation rather than wavelength-modulation: the pulse train at f_0 excites the QTF at resonance, and the photoacoustic signal is recovered at the fundamental (1f) with an EG&G 5110 lock-in amplifier connected to the ADM signal output. The lock-in reference is derived from the QCL TRIG OUT port, which emits a 5 V TTL level while the laser is firing.

Two practical issues were addressed. First, below a pulse width of ≈ 400 ns the lock-in fails to lock to the TRIG OUT reference (UNLK indicator). This is solved by using a second waveform-generator channel as a *dummy reference* with a fixed 500 ns pulse width, phase-matched to TRIG OUT (a phase offset of $\approx 217.4^\circ$ compensates the QCL internal-electronics delay). Second, at sensitivity settings of 500 μ V, 5 mV and 50 mV the analog output reads $\approx 40\%$ high for time constants ≥ 30 ms; these settings are therefore avoided (see Section 4.2).

3.5 Beam Delivery and Alignment

The QCL emits a collimated beam of waist $w_0 = 2.5$ mm. A gold-coated plane mirror steers the beam by 90° into a gold-coated off-axis parabolic (OAP) mirror (Thorlabs MPD149-M01, reflected focal length $f = 101.6$ mm), which focuses it into the ADM. For a Gaussian beam, the focused waist is

$$w'_0 \approx \frac{\lambda f}{\pi w_0} \approx 100 \mu\text{m}, \quad (3.1)$$

at $\lambda = 7.7 \mu\text{m}$, with a Rayleigh range $z_R \approx 4.1$ mm. The QTF is placed at the focus; at the entrance window (≈ 22.1 mm before the focus) the beam diameter is ≈ 1.1 mm, comfortably below the ≈ 1.6 mm clear aperture of the ADM.

Alignment proceeds by first making the beam parallel to the optical axis with the plane mirror, then setting the OAP at the focal distance and orienting it to inject the beam through the entrance window; the five-axis stage is then used to maximise the transmitted power at the output window. Comparing input and output power against the predicted window transmission (Section 3.6) confirms that the beam clears the prongs and resonator without clipping.

3.6 Optical Power Budget

The expected transmission through the BaF₂ windows is estimated by combining the Fresnel reflection at each interface with Beer–Lambert bulk absorption,

$$T_{\text{window}} = (1 - R)^2 e^{-\mu_a d}, \quad R = \left(\frac{n - 1}{n + 1} \right)^2, \quad (3.2)$$

where n is the BaF₂ refractive index from the Sellmeier relation. At $\lambda = 7.7 \mu\text{m}$, $n = 1.429$, giving $R \approx 0.0312$ and a single-interface transmission of $\approx 96.9\%$. The bulk absorption coefficient $\mu_a \approx 1.9 \times 10^{-3}/\text{mm}$ is obtained from the manufacturer's 92.07% transmission for a 10 mm sample. For the 2.5 mm ADM windows this yields $T_{\text{window}} \approx 93.4\%$ per window, and, applying Equation (3.2) to both the entrance and exit windows, $T_{\text{total}} \approx 87.2\%$ (a 12.8% loss). This wavelength-specific figure is used during alignment to distinguish genuine window loss from clipping inside the module.

3.7 Gas Handling and Concentration Control

The gas line uses two Alicat mass-flow controllers (MFCs): an MC-10SCCM-D (≤ 10 sccm) delivering the analyte (a certified 100 ppm N₂O in N₂ cylinder) and an MC-200SCCM-D/5M delivering N₂ buffer. Their outputs are combined and fed to an Alicat PCD-15PSIA pressure controller that regulates the ADM chamber pressure (one port to the closed ADM, one to a Vacuubrand MZ-2C-NT pump exhausting to the fume hood). Flows are set in the FlowVision v2.1.3 software. The delivered concentration is

$$C = c_0 \frac{F_{\text{N}_2\text{O}}}{F_{\text{N}_2\text{O}} + F_{\text{N}_2}}, \quad (3.3)$$

where $c_0 = 100$ ppm is the cylinder concentration and F are the flow rates; equal flows therefore give 50 ppm. Chamber pressure was typically 500 Torr to 600 Torr.

Chapter 4

Results and Discussion

Note: the values reported here are from preliminary characterization and detection runs (November 2025–March 2026) and are intended as a starting point; final, fully calibrated measurements should replace them before submission.

This chapter reports the experimental characterization of the sensor’s subsystems and the first N₂O detection results, followed by the identification of water-vapour interference, the observed pressure dependence, and a discussion of the current performance and its limitations.

4.1 Tuning-Fork / ADM Resonance Characterization

The QTF resonance was located by driving the ADM modulation input with a sinusoid from the waveform generator (100 mV peak-to-peak, 50 Ω) and recording the amplified output on the oscilloscope while sweeping the drive frequency in 0.1 Hz steps. Starting from the nominal 12 459.2 Hz, the response fell from ≈ 800 mV to ≈ 400 mV within 1 Hz; sweeping downwards, a clear maximum of 930 mV (1.85 V peak-to-peak) was found at 12 458.8 Hz. The operating resonance frequency was therefore fixed at

$$f_0 = 12\,458.8 \text{ Hz.} \quad (4.1)$$

This value is used as the QCL repetition (modulation) frequency throughout.

4.2 Lock-In Amplifier Characterization

The EG&G 5110 lock-in was characterized at f_0 by varying the modulation amplitude applied to the QTF and recording the demodulated magnitude R (Figure 4.1). The response is approximately linear in the input amplitude down to ≈ 4 mV, below which the oscilloscope trace loses its sinusoidal shape and R becomes unreliable.

A reproducible instrument artefact was also found: on the 500 μ V, 5 mV and 50 mV sensitivity ranges the analog output reads ≈ 40 % higher than expected when the time constant is ≥ 30 ms, in both XY and R – θ modes; reducing the time constant to ≤ 10 ms restores the expected value. All other ranges are consistent. These three ranges are therefore avoided, or used only with short time constants.

*Lock-in magnitude R vs. QTF modulation amplitude at f_0
(linear region above ≈ 4 mV). Replace with your plot.*

Figure 4.1. Lock-in amplifier response characterization at f_0 .

4.3 QCL Power and Tuning Characterization

The average output power of the MIRcat-2300 was measured across its tuning range for several pulse settings (Figure 4.2). The power varies strongly with wavelength, from below 1 mW near the band edges to a maximum of ≈ 124 mW around $7.5\text{ }\mu\text{m}$ to $7.6\text{ }\mu\text{m}$ (100 kHz, 200 ns). At the N_2O operating line ($1297.8/\text{cm}$, $\approx 7.7\text{ }\mu\text{m}$) the available power is ample, and the working point used for detection was ≈ 20 mW at 1000 ns pulse width. This tuning map also guides line selection, since regions of low output power (e.g. near $6.2\text{ }\mu\text{m}$) cannot be probed effectively.

*QCL average output power vs. wavelength over $5\text{ }\mu\text{m}$ to
 $9\text{ }\mu\text{m}$. Replace with your measured tuning curve.*

Figure 4.2. Measured QCL average output power across the tuning range.

4.4 Optical Power Budget Verification

After alignment, the input and output powers were compared against the predicted two-window transmission of 87.2 % (Section 3.6). At $\lambda = 7.7\text{ }\mu\text{m}$ with a 500 ns pulse width, $P_{\text{in}} = 3.85$ mW gave $P_{\text{out}} = 3.15$ mW; the window model predicts 3.35 mW, leaving ≈ 0.2 mW of additional loss inside the module. This small residual indicates the beam is essentially clearing the prongs and resonator, confirming acceptable alignment.

4.5 Detection of Nitrous Oxide

N_2O was first detected at 1297.8/cm with the QCL at 1000 ns / 20 mW, 100 ppm N_2O in a closed chamber at 500 Torr. The lock-in, in %FS $R-\theta$ mode at 20 mV sensitivity, showed a stable photoacoustic signal of ≈ 12 mV. Toggling the laser and the gas flow off and on removed the signal in every case, confirming that the reading is a genuine photoacoustic response to N_2O and not an electrical or background artefact.

Two N_2O lines were then compared (800 ns, 100 ppm, 600 Torr): 1297.8/cm (22 mW) and 1271.5/cm (19 mW) gave 14.0 mV and 15.5 mV respectively (Table 4.1).

Table 4.1. Photoacoustic signal at two N_2O lines (100 ppm, 600 Torr, 800 ns).

Line (1/cm)	QCL power (mW)	Signal (mV)
1297.8	22	14.0
1271.5	19	15.5

4.6 Spectral Interference from Water Vapour

When the laser was tuned to a nominally non-absorbing region near 1400/cm as a control, strong signals were nonetheless observed. Scanning the QCL revealed peaks at 1456.9, 1405.2, 1288.5 and 1287.5/cm (15.0, 29.0, 23.0 and 28.3 mV), and further peaks near 1798.5 and 1772.7/cm (16 and 20 mV). Cross-referencing with HITRAN [3] attributes these to water vapour (H_2O) present in the chamber; nitrogen shows no absorption in this window. Water-vapour interference is thus a significant practical issue for N_2O detection in this spectral region (Figure 4.3) and motivates either gas drying or careful line selection away from H_2O features.

H_2O absorption over 1111/cm to 2000/cm (HITRAN), with the observed interference peaks marked. Replace with your figure.

Figure 4.3. Water-vapour absorption spectrum from HITRAN in the QCL tuning window.

4.7 Pressure Dependence

The photoacoustic signal increased approximately linearly with chamber pressure over the range explored. Purging with N_2 while pumping reduced the pressure to 60 Torr to 70 Torr and brought the lock-in reading to essentially zero; whether this is due to the low pressure itself or to the removal of residual absorbers (including H_2O) was not separated and remains an open question. **[TODO: Add a controlled signal-vs-pressure measurement at fixed concentration.]**

4.8 Discussion

These results demonstrate a working pulsed-QCL QEPAS sensor that detects N_2O at the 50 ppm to 100 ppm level and a complete, characterized detection chain. A background measurement highlights the main current limitation: at the non-absorbing line 1340/cm the lock-in read 1.08 mV (3.18 mW), rising only to 1.52 mV at 1297.8/cm (3.23 mW). The large wavelength-independent background — despite an in-module loss of only ≈ 0.2 mW — points to residual beam clipping / wall heating from imperfect alignment rather than window loss, and currently limits the achievable detection limit.

The remaining work is to suppress this background (alignment, gas drying), and then to convert the measured signals into quantitative figures of merit: a concentration calibration curve, the normalized noise-equivalent absorption (NNEA), the minimum detectable concentration, and an Allan-deviation analysis of the optimal integration time. Water-vapour interference and the pressure dependence should be characterized systematically as part of this.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

Summarize what was accomplished: the design and construction of a pulsed-QCL QEPAS sensor, the characterization of its tuning fork / ADM, lock-in chain, and QCL, the optical power budget, and the validated detection of N₂O with its measured sensitivity and stability. Restate the headline performance numbers and why they matter.

5.2 Future Work

Outline the next steps. The primary direction is extending the platform from N₂O to the detection of trace volatile organic compounds (VOCs), such as plant-emitted VOCs, with the goal of a research publication. Discuss the improvements this may require (e.g. line selection for the target VOC, higher sensitivity, interference handling, and calibration).

References

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Appendix A

Supplementary Material

Use appendices for material that supports the thesis but would interrupt the flow of the main text: long derivations, extra plots, code listings, datasheets, or detailed tables.

Each `\chapter` after `\appendix` is lettered automatically (Appendix A, Appendix B, ...).